

Dynamics and Reconstruction in the Observable Measure

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Abstract

A σ -additive probability measure on an observable algebra [12] already determines, via the Rokhlin disintegration theorem, a disintegration kernel for any pair of observables and a minimal sufficient factor carrying full conditional information. When the observables are indexed over time, temporal coherence forces the Chapman–Kolmogorov equation; a Markov semigroup and Koopman–Perron duality follow without a separately posited dynamical law. In a measure-preserving system the minimal sufficient factor becomes the delay map, and reconstruction — that the observable σ -algebra equals the full σ -algebra modulo null sets, that the delay algebra is dense in L^2 , and that the delay map is a measure-theoretic embedding — reduces to the faithfulness of this factor. The Stone space built in [12] to realise σ -additivity is then identified with the state space itself.

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1 Introduction

Koopman studied dynamical systems through observables [9], but the dynamical law T remains primitive: the operator $U_T f = f \circ T$ is defined only after an underlying evolution has already been specified. The classical stochastic-process tradition proceeds similarly. Prediction is formulated relative to a filtration, so temporal order and information flow are built into the formalism from the outset [4]. That ordering is natural, but it need not be conceptually prior. One may instead ask whether observable probability already carries a conditional regularity between observables, and whether the familiar semigroup and predictive structures arise only when that regularity is organised coherently over time.

In [12], σ -additivity is the irreducible commitment that turns coherent charges into a probability measure: no first-order condition can force it, and compatibility collapses every candidate reformulation back to σ -additivity itself. The Stone construction produces a measure unconditionally, but only σ -additivity ensures it lives on the space of realised states. Once committed to, however, the measure already carries more than it was built to carry. Any two observations — no hypothesis on time or order — determine a conditional regularity: the Rokhlin disintegration theorem [13] forces a Markov kernel into existence, from which a minimal sufficient factor emerges as the coarsest reduction of the sample point carrying full conditional information.

Index this regularity over time and the kernels must compose consistently. That coherence forces the Chapman–Kolmogorov equation, yielding a Markov semigroup and Koopman–Perron duality without a separately posited dynamical law. In a measure-preserving system the minimal sufficient factor becomes the delay map, and the question becomes internal: does the observable algebra generated by the delay orbit equal the full σ -algebra? The reconstruction theorem gives three equivalent formulations; under any of them, the Stone space [12] built to realise σ -additivity collapses onto the state space itself.

2 Disintegration, dynamics, and duality

Let $(\Omega, \mathcal{F}, P)$ be a σ -additive probability space on an observable algebra [12], and let Q, F be two measurable maps:

- $Q : \Omega \rightarrow \alpha$, where $(\alpha, \mathcal{A})$ is a measurable space;
- $F : \Omega \rightarrow \beta$, where $(\beta, \mathcal{B})$ is a standard Borel space.

The standard Borel hypothesis on β is used once: it is the hypothesis under which the Rokhlin disintegration theorem [13] guarantees the existence of a regular conditional distribution. All other results hold in greater generality.

We write $\mathcal{P}(\beta)$ for the space of probability measures on β , equipped with its canonical (Giry) σ -algebra. The joint pushforward $\rho := P \circ (Q, F)^{-1}$ on $\alpha \times \beta$ has marginals $\rho_\alpha = P \circ Q^{-1}$ and $\rho_\beta = P \circ F^{-1}$.

The regular conditional distribution of F given Q is the unique (up to $P \circ Q^{-1}$ -null sets) Markov kernel $\kappa_Q : \alpha \rightarrow \mathcal{P}(\beta)$ satisfying

$$\int_A \kappa_Q(q, B) d(P \circ Q^{-1})(q) = P(Q \in A, F \in B)$$

for all measurable $A \subseteq \alpha$ and $B \subseteq \beta$.

Definition 2.1 (Disintegration kernel). The *disintegration kernel* of F given Q is κ_Q , the regular conditional distribution of $\rho = P \circ (Q, F)^{-1}$ over its first marginal: $\kappa_Q(q, A) = P(F \in A \mid Q = q)$.

The kernel is covariant under measurable maps: if $\pi : \alpha \rightarrow \alpha'$ is measurable and $Q' = \pi \circ Q$, then $\kappa_Q(q, \cdot) = \kappa_{Q'}(\pi(q), \cdot)$ for $(P \circ Q^{-1})$ -almost every q , by uniqueness of disintegration.

Definition 2.2 (Minimal sufficient factor). The *minimal sufficient factor* is

$$Q_* : \Omega \rightarrow \mathcal{P}(\beta), \quad Q_*(\omega) = \kappa_Q(Q(\omega), \cdot).$$

Since κ_Q is a Markov kernel it is measurable, and Q_* inherits measurability by composition.

Theorem 2.3 (Factorization via the canonical factor). *For every bounded measurable $g : \beta \rightarrow \mathbb{R}$:*

$$E[g(F) \mid \sigma(Q)] = \int g dQ_*(\cdot) \quad P\text{-a.e.}$$

Proof. Let $h(\omega) := \int g dQ_*(\omega)$. We verify the two defining properties.

Measurability. The map $q \mapsto \int g d\kappa_Q(q, \cdot)$ is strongly measurable; since $Q_* = \kappa_Q(Q(\cdot), \cdot)$, the map h is $\sigma(Q)$ -measurable.

Set-integral identity. For $S = Q^{-1}(A)$:

$$\begin{aligned} \int_S h dP &= \int_A \int g d\kappa_Q(q, \cdot) d(P \circ Q^{-1})(q) \quad (\text{pushforward}) \\ &= \int_{A \times \beta} g(f) d\rho(q, f) \quad (\text{disintegration}) \\ &= \int_S g(F) dP. \end{aligned}$$

The conclusion follows from a.e. uniqueness of conditional expectation. □

Since $\sigma(Q_*) \subseteq \sigma(Q)$ and the right-hand side is already $\sigma(Q_*)$ -measurable, Q_* is a sufficient statistic for F given Q : $E[g(F) \mid \sigma(Q)] = E[g(F) \mid \sigma(Q_*)]$ P -a.e. It is *minimal* in the sense that every sufficient statistic factors through it: Q_* is the coarsest reduction of ω that carries the full conditional regularity of F given Q . The operator $(K_{Q_*}g)(q_*) = \int g dq_*$ on $\mathcal{P}(\beta)$ recasts this as $E[g(F) \mid \sigma(Q)] = (K_{Q_*}g) \circ Q_*$ P -a.e.

When the disintegration kernel κ_Q is indexed over a time parameter, the kernels cannot be chosen independently: the regularity of F at time $t + s$ given Q at time 0 must be consistent with first observing at time s and then at time t . This consistency requirement forces the Chapman–Kolmogorov equation.

Concretely, index over a discrete time horizon $\mathbb{N}$ and let $\Pi_t : \gamma \rightarrow \mathcal{P}(\gamma)$ be a Markov kernel on a state space γ , with $\Pi_0(q, \cdot) = \delta_q$. Temporal coherence of the conditional regularity demands

$$\Pi_{t+s}(q, \cdot) = \int_{\gamma} \Pi_t(q', \cdot) d\Pi_s(q, q') \quad (\text{i.e., } \Pi_{t+s} = \Pi_t \circ_{\kappa} \Pi_s)$$

— the Chapman–Kolmogorov equation, derived from the structure rather than assumed. A family $\{\Pi_t\}$ satisfying it is a *Markov semigroup kernel*. The associated *Markov operators* are

$$(K_t g)(q) := \int g d\Pi_t(q, \cdot), \quad g : \gamma \rightarrow \mathbb{R}.$$

Each K_t is positive, unital, and contractive on bounded measurable functions, with $K_0 = \text{id}$; positivity, unitality, and contractiveness are immediate from the Markov property.

Proposition 2.4 (Semigroup property). $K_{t+s}g = K_t(K_s g)$ for all bounded measurable g .

Proof. $(K_{t+s}g)(q) = \int g d(\Pi_t \circ_{\kappa} \Pi_s)(q, \cdot) = \int (\int g d\Pi_t(q', \cdot)) d\Pi_s(q, q') = \int (K_t g)(q') d\Pi_s(q, q') = (K_t(K_s g))(q)$, where the exchange of integrals is justified by Bochner–Fubini. $\square$

The pushforward of a measure μ on γ through the kernel is $P_t^* \mu := \mu \star \Pi_t = \int \Pi_t(q, \cdot) d\mu(q)$.

Theorem 2.5 (Koopman–Perron duality). For every bounded measurable g and finite measure μ :

$$\int (K_t g) d\mu = \int g d(P_t^* \mu).$$

Proof. $\int g d(P_t^* \mu) = \int g d(\mu \star \Pi_t) = \int_{\gamma} \int g d\Pi_t(q, \cdot) d\mu(q) = \int (K_t g) d\mu$, by Bochner–Fubini. $\square$

When $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$ for a measurable map $\varphi_t : \gamma \rightarrow \gamma$, the stochastic picture collapses.

Proposition 2.6 (Deterministic limit). If $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$, then $K_t g = g \circ \varphi_t$ and $\varphi_{t+s} = \varphi_t \circ \varphi_s$.

Proof. $(K_t g)(q) = \int g d\delta_{\varphi_t(q)} = g(\varphi_t(q))$. The flow law follows from $\delta_{\varphi_{t+s}(q)} = (\Pi_t \circ_{\kappa} \Pi_s)(q, \cdot) = \delta_{\varphi_t(\varphi_s(q))}$ and injectivity of the Dirac embedding. $\square$

In the deterministic case, $K_t g = g \circ \varphi_t = U_T^t g$: the Markov operator specialises to the classical Koopman operator. The Koopman–Perron duality $\int K_t g d\mu = \int g d(P_t^* \mu)$ becomes the change-of-variables formula $\int g \circ T^t d\mu = \int g d(T_*^t \mu)$.

Example 2.7 (Finite Markov chain). Let $X = \{1, \dots, k\}$, $\mathcal{B} = 2^X$, and let $P = (p_{ij})$ be the transition matrix of a finite irreducible Markov chain with unique stationary distribution μ ($\mu P = \mu$). Take full-state observation $Q_t = X_t$. The disintegration kernel of X_1 given X_0 is $\kappa(i, \cdot) = P(i, \cdot)$, the i -th row of P : this is the unique Markov kernel satisfying $\mathbb{E}_{\mu}[f(X_1) \mid X_0 = i] = \sum_j p_{ij} f(j)$ for all bounded f . The t -step kernel $\Pi_t(i, \cdot) = P^t(i, \cdot)$ is the corresponding semigroup kernel, and temporal coherence gives $\Pi_{t+s} = \Pi_t \circ_{\kappa} \Pi_s$, i.e. $P^{t+s} = P^t P^s$. The semigroup acts as

$$K_t f(i) = \sum_j p_{ij}^t f(j).$$

Koopman–Perron duality reduces to the stationary-measure identity:

$$\int K_t g \, d\mu = \mu^\top P^t g = (\mu P^t)^\top g = \mu^\top g = \int g \, d\mu,$$

the fixed-point property $\mu P = \mu$ of the stationary distribution. In this finite setting every charge is automatically σ -additive, so the admissibility condition of [12] is trivially satisfied. Reconstruction holds at lag 0: $\mathcal{O}_h = \sigma(h) = \mathcal{B}(X)$ whenever h separates points of X .

The full structure produced by the construction is the triple $(\mathcal{P}(\beta), \nu_{Q_*}, \{K_t\}_{t \in \mathbb{N}})$: the canonical factor space, stationary distribution $\nu_{Q_*} = P \circ Q_*^{-1}$, and Markov semigroup. This is the *observable dynamical system* derived from (Q, F, P) : the conditional regularity of F relative to Q , promoted to a dynamical object by temporal coherence.

3 Reconstruction

The structures of §2 require only a probability measure and two observables. We now specialise: let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system and $h \in L^\infty$ an observable. Setting $Q_t = h \circ T^t$ gives $Q_*(\omega) = (h(T^n \omega))_{n \in \mathbb{Z}}$, the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$, and the observable dynamical system $(\Phi_h(X), \mu \circ \Phi_h^{-1}, \{K_t\})$. The measure-preserving hypothesis and the choice of h are genuine inputs; what follows from them is not.

The canonical factor space $\Phi_h(X)$ is a subset of $\mathbb{R}^{\mathbb{Z}}$ encoding all dynamical information accessible to h . Whether it is a faithful image of X depends on whether the delay orbit $\{h \circ T^n : n \in \mathbb{Z}\}$ separates points of X . Injectivity of Φ_h modulo μ -null sets is equivalent to the observable σ -algebra

$$\mathcal{O}_h = \sigma(\{h \circ T^n : n \in \mathbb{Z}\})$$

equalling $\mathcal{B}(X)$ modulo μ .

Definition 3.1 (Faithful factor). A measurable map $\psi : (X, \mathcal{B}, \mu) \rightarrow (Y, \mathcal{Y})$ is *faithful modulo μ* if

$$\sigma(\psi) = \mathcal{B} \quad \text{modulo } \mu,$$

equivalently, if ψ separates μ -a.e. pair of points of X . The reconstruction question is whether Φ_h is faithful modulo μ .

The Koopman–Perron duality (Theorem 2.5) connects this to a function-space question: the observable σ -algebra $\mathcal{O}_h$ determines the Markov semigroup through the algebra it generates, and the reconstruction question is precisely whether that algebra is large enough to recover all of $\mathcal{B}$.

For a finite joined query algebra $\mathcal{G} = \mathcal{E}_{n_1} \vee \cdots \vee \mathcal{E}_{n_k}$ from a directed system $(\mathcal{E}_n)_n$, the *separation defect* is

$$\delta(\mathcal{G}) := \sup_{A \in \sigma(\mathcal{C})} \inf_{E \in \mathcal{G}} \mu(A \triangle E),$$

where $\sigma(\mathcal{C})$ is the observable σ -algebra of the full system. It vanishes iff $\sigma(\mathcal{G}) = \sigma(\mathcal{C}) \bmod \mu$. A sequence $(\mathcal{G}_k)$ is *honest* if it is increasing and $\sigma(\bigcup_k \mathcal{G}_k) = \sigma(\mathcal{C}) \bmod \mu$; under an honest scheme $\delta(\mathcal{G}_k)$ is non-increasing.

Theorem 3.2 (CE drives separation defect to zero). *Let $(\mathcal{E}_n)_n$ be a query system with realization measure μ satisfying collective exhaustion [12]. Under any honest refinement scheme $(\mathcal{G}_k)$,*

$$\delta(\mathcal{G}_k) \longrightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Proof. Fix $A \in \sigma(\mathcal{C})$. Since $(\mathcal{G}_k)$ is honest, $\sigma(\bigcup_k \mathcal{G}_k) = \sigma(\mathcal{C}) \bmod \mu$, so A is $\sigma(\bigcup_k \mathcal{G}_k)$ -measurable. By the $L^1(\mu)$ martingale convergence theorem, $\mathbf{1}_{E_k} \rightarrow \mathbf{1}_A$ in $L^1(\mu)$ where $E_k = \mathbb{E}_\mu[\mathbf{1}_A \mid \mathcal{G}_k]$ rounded to the nearest atom; in particular $\mu(A \triangle E_k) \rightarrow 0$. CE [12] ensures no mass persists on the vanishing sets $A \triangle E_k$, confirming $\inf_{E \in \mathcal{G}_k} \mu(A \triangle E) \rightarrow 0$ uniformly over A . Hence $\delta(\mathcal{G}_k) \rightarrow 0$. $\square$

Remark 3.3 (The temporal specialisation). With $Q_t = h \circ T^t$, the algebra $\mathcal{O}_h^{(L)}$ is an honest refinement scheme once reconstruction holds. The separation defect specialises to $\delta(L) := \delta(\mathcal{O}_h^{(L)})$; under mixing, $\delta(L)$ decays exponentially. In the general query setting, exhaustion and mixing are separate conditions; the delay-map setting collapses the first by construction.

Throughout, $(X, \mathcal{B}, \mu)$ is a standard probability space, $T : X \rightarrow X$ is a measurable, μ -preserving invertible bimeasurable bijection with $T_*\mu = \mu$, and $h \in L^\infty(X, \mu)$.

Definition 3.4 (Observable algebra). The *observable algebra* generated by h is

$$\mathcal{O}_h := \sigma(\{h \circ T^n : n \in \mathbb{Z}\}) \subseteq \mathcal{B}(X).$$

The *delay algebra* $\mathcal{A}_h$ is the smallest subalgebra of $L^\infty(X, \mu)$ containing 1 and every $h \circ T^n$; concretely, finite sums $\sum_\alpha c_\alpha \prod_k (h \circ T^{n_k})^{e_k}$.

Definition 3.5 (Delay map). The *delay map* is

$$\Phi_h : X \rightarrow \mathbb{R}^{\mathbb{Z}}, \quad \Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}.$$

Since $\pi_n \circ \Phi_h = h \circ T^n$ for each coordinate projection π_n , the map Φ_h is measurable and $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$. Thus $\mathcal{O}_h = \mathcal{B} \bmod \mu$ is precisely the condition that Φ_h is a measure-theoretic embedding.

The central lemma connects the Boolean algebra level to the L^2 level. It holds for any subalgebra of L^∞ .

Lemma 3.6 (Density bridge). *Let $(X, \mathcal{B}, \mu)$ be a probability space and let $\mathcal{A} \subseteq L^\infty(X, \mu)$ be a subalgebra containing the constant function 1. Then:*

$$\overline{\mathcal{A}}^{L^2} = L^2(X, \sigma(\mathcal{A}), \mu).$$

Consequently, $\overline{\mathcal{A}}^{L^2} = L^2(X, \mu)$ if and only if $\sigma(\mathcal{A}) = \mathcal{B}$ modulo μ .

Proof. Write $\mathcal{M} := \sigma(\mathcal{A})$ and $V := \overline{\mathcal{A}}^{L^2}$.

Step 1: $V \subseteq L^2(X, \mathcal{M}, \mu)$. Every $f \in \mathcal{A}$ is $\mathcal{A}$ -measurable, hence $\mathcal{M}$ -measurable. So $\mathcal{A} \subseteq L^2(X, \mathcal{M}, \mu)$. Since $L^2(X, \mathcal{M}, \mu)$ is a closed subspace of $L^2(X, \mathcal{B}, \mu)$, taking the closure gives $V \subseteq L^2(X, \mathcal{M}, \mu)$.

Step 2: $L^2(X, \mathcal{M}, \mu) \subseteq V$. It suffices to show that every indicator $\mathbf{1}_E$ with $E \in \mathcal{M}$ lies in V . By the functional monotone class theorem [5], the class

$$\mathcal{H} := \{f \in L^\infty(X, \mathcal{M}, \mu) : f \in V\}$$

is a vector space containing $\mathcal{A}$ and closed under bounded monotone pointwise limits (since if $f_n \in V$ are uniformly bounded and $f_n \rightarrow f$ pointwise a.e. with $f \in L^\infty$, then $f_n \rightarrow f$ in L^2 by dominated convergence, so $f \in V$). Since $\mathcal{A}$ is an algebra containing 1 and $\sigma(\mathcal{A}) = \mathcal{M}$, the monotone class theorem gives $\mathcal{H} = L^\infty(X, \mathcal{M}, \mu)$. In particular every $\mathcal{M}$ -indicator $\mathbf{1}_E \in V$.

Since the $\mathcal{M}$ -simple functions are dense in $L^2(X, \mathcal{M}, \mu)$ and V is closed, $L^2(X, \mathcal{M}, \mu) \subseteq V$.

Conclusion. Steps 1 and 2 give $V = L^2(X, \mathcal{M}, \mu)$. The final equivalence is immediate: $V = L^2(X, \mathcal{B}, \mu)$ iff $L^2(X, \mathcal{M}, \mu) = L^2(X, \mathcal{B}, \mu)$ iff $\mathcal{M} = \mathcal{B} \bmod \mu$. $\square$

Remark 3.7 (Role of the algebra structure). The algebra structure of $\mathcal{A}$ is essential. The corresponding statement for the closed *linear span* $\mathcal{H}_{\mathcal{A}} = \overline{\text{span}}(\mathcal{A})$ is false in general: $\mathcal{H}_{\mathcal{A}} = L^2$ does not imply $\sigma(\mathcal{A}) = \mathcal{B}$.

For example, take $X = [0, 1]$, $\mu = \text{Lebesgue}$, $T(x) = 2x \bmod 1$ (the doubling map), and $h(x) = x$. The orbit $\{h \circ T^n\}(x) = \{2^n x \bmod 1\}$ generates $\mathcal{B}([0, 1])$ as a σ -algebra (the dyadic structure separates points). But $\text{span}\{2^n x \bmod 1 : n \geq 0\}$ is a proper subspace of $L^2[0, 1]$: it does not contain x^2 . So the linear span can be a proper dense subspace even when $\sigma(\mathcal{A}) = \mathcal{B}$.

In the other direction, h cyclic for U_T (linear span dense) always implies $\sigma(\mathcal{A}_h) = \mathcal{B} \bmod \mu$ — see Corollary 3.10.

Theorem 3.8 (Reconstruction theorem). *Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system and $h \in L^\infty(X, \mu)$. The following are equivalent:*

- (i) $\mathcal{O}_h = \mathcal{B} \bmod \mu$;
- (ii) $\mathcal{A}_h$ is dense in $L^2(X, \mu)$;
- (iii) $\Phi_h : X \rightarrow \mathbb{R}^{\mathbb{Z}}$ is a measure-theoretic embedding: $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B} \bmod \mu$.

Proof. (i) $\Leftrightarrow$ (ii). Apply the density bridge (Lemma 3.6) with $\mathcal{A} = \mathcal{A}_h$ and $\sigma(\mathcal{A}_h) = \mathcal{O}_h$: $\overline{\mathcal{A}_h}^{L^2} = L^2(X, \mu)$ iff $\mathcal{O}_h = \mathcal{B} \bmod \mu$.

(i) $\Leftrightarrow$ (iii). By Definition 3.5, $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$. So $\mathcal{O}_h = \mathcal{B} \bmod \mu$ iff $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B} \bmod \mu$, which is exactly condition (iii). $\square$

Condition (iii) means Φ_h induces an isomorphism of measure algebras $(X/\mu, \mathcal{B}/\mu) \cong (\mathbb{R}^{\mathbb{Z}}/(\Phi_h)_*\mu, \mathcal{B}(\mathbb{R}^{\mathbb{Z}})/(\Phi_h)_*\mu)$. Since $(\Phi_h(Tx))_n = h(T^{n+1}x) = (\sigma(\Phi_h(x)))_n$, the measure $(\Phi_h)_*\mu$ is shift-invariant, and (X, T, μ) is isomorphic to $(\mathbb{R}^{\mathbb{Z}}, \sigma, (\Phi_h)_*\mu)$ restricted to $\Phi_h(X)$.

Reconstruction is an asymptotic condition: the finite-lag algebras $\mathcal{O}_h^{(L)} \uparrow \mathcal{O}_h$ are monotone, and the separation defect $\delta(L) = \delta(\mathcal{O}_h^{(L)})$ measures the gap at finite lag. The rate at which $\delta(L)$ vanishes determines the practical content of the theorem and depends on mixing and geometric hypotheses beyond the scope of the present paper.

Definition 3.9 (Cyclic vector). The function $h \in L^2(X, \mu)$ is a *cyclic vector* for U_T if

$$\mathcal{H}_h := \overline{\text{span}}\{U_T^n h : n \in \mathbb{Z}\} = L^2(X, \mu).$$

Corollary 3.10 (Cyclic implies reconstruction). *If $h \in L^\infty(X, \mu)$ is a cyclic vector for U_T , then $\mathcal{O}_h = \mathcal{B} \bmod \mu$.*

Proof. Since $\mathcal{A}_h$ contains $\text{span}\{h \circ T^n\} = \text{span}\{U_T^n h\}$ as a subset, $\overline{\mathcal{A}_h}^{L^2} \supseteq \mathcal{H}_h = L^2$. By the density bridge, $\mathcal{O}_h = \mathcal{B} \bmod \mu$. $\square$

Remark 3.11 (The implication is strict). The converse of Corollary 3.10 fails in general: $\mathcal{O}_h = \mathcal{B}$ does not imply h is cyclic. The doubling map example of Remark 3.7 witnesses this.

The cyclic vector condition asks for density of the *linear span* of the orbit; reconstruction asks only for density of the *algebra*. For a bounded h , the linear span is contained in the algebra, so cyclicity is the stronger condition.

The two conditions coincide precisely when U_T has *simple L^2 -spectrum*: $L^2(X, \mu)$ is a cyclic U_T -module, meaning there exists some cyclic vector. In this case, every generating observation h (satisfying $\mathcal{O}_h = \mathcal{B}$) is automatically cyclic. Simple spectrum is a generic property of ergodic transformations in the Baire category sense [6] but is not universal.

The Stone space $\text{St}(\mathcal{O}_h)$ of the observable algebra is the compact Hausdorff space whose points are ultrafilters on $\mathcal{O}_h$, with the natural Stone embedding $\iota : X \rightarrow \text{St}(\mathcal{O}_h)$ sending x to the principal ultrafilter $\{E \in \mathcal{O}_h : x \in E\}$.

Theorem 3.12 (Stone space identification). *Suppose $\mathcal{O}_h = \mathcal{B}$ modulo μ . Then ι is a measure-theoretic isomorphism: ι identifies the measure algebra $(X/\mu, \mathcal{B}/\mu)$ with $(\text{St}(\mathcal{O}_h)/\iota_*\mu, \mathcal{B}(\text{St}(\mathcal{O}_h))/\iota_*\mu)$.*

Proof. Injectivity mod μ . Suppose $\iota(x) = \iota(x')$. For any $E \in \mathcal{B}$ with $x \in E \not\equiv x'$, find $F \in \mathcal{O}_h$ with $\mu(E \Delta F) = 0$. Since $x \in F$ and $\iota(x) = \iota(x')$, also $x' \in F$, so $x' \in (E \Delta F) \cup F^c$, placing x' in a null set. Hence ι is injective μ -a.e.

σ -algebra identification. The pullback ι^{-1} maps the Baire σ -algebra of $\text{St}(\mathcal{O}_h)$ (generated by the clopen sets $[E] = \{u : E \in u\}$) to $\mathcal{O}_h = \mathcal{B} \bmod \mu$.

Measure recovery. Since μ is σ -additive on $\mathcal{O}_h = \mathcal{B}$, the collective exhaustion condition of [12] forces $\iota_*\mu$ to concentrate on the principal ultrafilters $\iota(X)$, so $\iota_*\mu$ is supported on the image and the measure algebras are identified. $\square$

The Stone space [12] — built as a compact ambient space for realising σ -additivity — is, under the reconstruction condition, the state space X itself. The construction that began by seeking a compact completion of the sample space ends by recognising it as the object the observer was trying to recover.

Example 3.13 (Cantor set: reconstruction without smoothness). Let $\mathcal{C} \subset [0, 1]$ be the middle-thirds Cantor set, and let $\mu_{\mathcal{C}}$ be the Cantor measure — the pushforward of the fair-coin product measure $\nu = (\frac{1}{2}\delta_0 + \frac{1}{2}\delta_2)^{\mathbb{N}}$ under the ternary coding map $\pi(d_1, d_2, \dots) = \sum_{k \geq 1} d_k/3^k$. Let $S : \mathcal{C} \rightarrow \mathcal{C}$ be the shift map

$$S(x) = \begin{cases} 3x & x \in \mathcal{C} \cap [0, \frac{1}{3}], \\ 3x - 2 & x \in \mathcal{C} \cap [\frac{2}{3}, 1], \end{cases}$$

and let $h(x) = x$. The system $(\mathcal{C}, \mu_{\mathcal{C}}, S)$ is measurably conjugate to the Bernoulli shift $(\{0, 2\}^{\mathbb{N}}, \nu, \sigma)$ via π ; in particular S is $\mu_{\mathcal{C}}$ -invariant and ergodic.

For $x = \pi(d_1, d_2, \dots)$ we have $S^n x = \pi(d_{n+1}, d_{n+2}, \dots)$. Since every point of $\mathcal{C}$ lies in $[0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$,

$$d_{n+1} = 0 \iff S^n x \in [0, \frac{1}{3}], \quad d_{n+1} = 2 \iff S^n x \in [\frac{2}{3}, 1],$$

so $h \circ S^n$ recovers the $(n+1)$ -st ternary digit of x . The delay vector $\Phi_h^{(L)}(x) = (x, Sx, \dots, S^L x)$ reads off digits $d_1, \dots, d_{L+1}$; its fibres are the generation- $(L+1)$ cylinder sets

$$F_z = \{x \in \mathcal{C} : d_k(x) = z_k, 1 \leq k \leq L+1\}, \quad z \in \{0, 2\}^{L+1},$$

each carrying $\mu_{\mathcal{C}}(F_z) = 2^{-(L+1)}$, so $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)}) = \sigma\{F_z : z \in \{0, 2\}^{L+1}\}$. Since $\mu_{\mathcal{C}}$ -a.e. pair of distinct points differs at some finite digit, these algebras separate points as $L \rightarrow \infty$:

$$\mathcal{O}_h = \sigma\left(\bigcup_{L \geq 0} \mathcal{O}_h^{(L)}\right) = \mathcal{B}(\mathcal{C}) \bmod \mu_{\mathcal{C}}.$$

By Theorem 3.8, Φ_h is a measure-theoretic embedding: the finite-lag observable algebras refine to the full Borel structure as $L \rightarrow \infty$.

The observable algebra also encodes the geometry of $\mathcal{C}$ directly. At each lag L there are 2^{L+1} fibres of Euclidean diameter $(1/3)^{L+1}$, so the box-counting ratio $\log 2^{L+1} / \log 3^{L+1} = \log 2 / \log 3$ is an algebraic invariant of the algebra itself, constant across all lags. The Hausdorff dimension is not a separate geometric calculation: it is read off from the refinement rate of the observable algebra.

$\mathcal{C}$ is not a smooth manifold: it is compact and zero-dimensional, with Hausdorff dimension $\log 2 / \log 3 \approx 0.63$. The differential-geometric hypotheses of Takens's theorem — C^2 ambient dynamics, integer manifold dimension d , embedding bound $2d+1$ — do not apply; nor do the finite-lag extensions of Sauer–Yorke–Casdagli [14], which still require a finite-dimensional ambient manifold or attractor, apply. Theorem 3.8 applies because it requires only that the observable algebra be measurably generated; smoothness and finite dimensionality play no role.

The classical Takens embedding theorem [15] reaches a related conclusion by a different route: for a C^2 diffeomorphism on a compact d -manifold and a generic smooth h , the finite-delay map $\Phi_h^{(k)}(x) = (h(x), h(Tx), \dots, h(T^{k-1}x))$ is a diffeomorphic embedding into $\mathbb{R}^{2d+1}$ for $k \geq 2d + 1$. Sauer–Yorke–Casdagli [14] extend this to prevalence-generic observables and Lipschitz maps, while retaining the finite-dimensional manifold hypothesis and the finite-lag, finite-dimension embedding framework. Botvinick–Greenhouse, Lunsford, and Wang [3] develop a measure-theoretic framework for delay embedding via optimal transport, replacing topological with distributional equivalence. The reconstruction theorem here takes a different route: it requires only measurability, uses all delays simultaneously, and replaces the dimension count with the algebraic density condition; the embedding is measure-theoretic rather than topological. On a compact manifold with generic smooth h , the orbit $\{h \circ T^n\}$ separates points, so Stone–Weierstrass gives $\mathcal{A}_h$ uniformly dense in $C(M)$ and hence L^2 -dense: the density condition holds and the results are compatible. For non-smooth or zero-dimensional systems — the Cantor example being representative — the closer historical ancestor is the symbolic dynamics and generating-partition tradition [1]: representing orbits through successively finer finite partitions is exactly what the observable algebra $\mathcal{O}_h^{(L)}$ does as L grows, with faithfulness of Φ_h replacing the generating-partition condition.

Remark 3.14 (Observable separation and Lyapunov exponents). When T is C^r and reconstruction holds, the delay map Φ_h carries a natural instability rate: for μ -a.e. x and nearby y ,

$$\lambda_h(x, y) := \limsup_{n \rightarrow \infty} \frac{1}{n} \log d(\Phi_h(T^n x), \Phi_h(T^n y)).$$

This *observable separation exponent* measures how fast the reconstructed dynamics distinguishes initial conditions through the observable, rather than how fast tangent vectors grow. When h is generic in the sense of Takens [15], the finite-lag embedding $\Phi_h^{(L)}$ is bi-Lipschitz on the attractor, sandwiching λ_h between the top Lyapunov exponent λ_1 from above and below; one expects $\lambda_h = \lambda_1$ μ -a.e., but making this precise requires comparing the induced metric on $\Phi_h^{(L)}(X)$ with the ambient Riemannian geometry as $L \rightarrow \infty$.

4 Discussion

The measure constructed in [12] already contains, without further assumption, the structures developed here. The Rokhlin disintegration reads off a disintegration kernel for any pair of observables; temporal indexing of that kernel forces Chapman–Kolmogorov and hence the Markov semigroup; and in a measure-preserving system the minimal sufficient factor becomes the delay map. Nothing is added to the measure — only consequences are drawn.

The mathematical content of §§2–3 is standard once a σ -additive measure is in hand on the observable algebra; what the programme changes is the explanatory position of each object. The individual components are classical. Rokhlin disintegration [13] gives regular conditional distributions on standard Borel spaces; Halmos and Savage [7] laid the measure-theoretic foundation for sufficient statistics via the Radon–Nikodym theorem; Blackwell [2] placed sufficiency in comparative terms; and Doob [4] established conditional expectation, filtrations, and Markov semigroups as the language of modern probability. Predictive-state representations [10] and observable operator models [8] likewise distrust hidden-state parametrisations and encode state through observable predictions. Here the kernel, the factor, the semigroup, and the delay map are not assumed, chosen, or built from a prior dynamical law — they are consequences of a single measure on an observable algebra, disclosed in sequence.

The reconstruction theorem closes a circle opened in [12]. The Stone space $\text{St}(\mathcal{C})$ built there to realise σ -additivity is an abstract completion of the observable algebra, containing the realised state space as the principal ultrafilters but not yet identified with it. When Φ_h is

faithful modulo μ (Definition 3.1), the three equivalent conditions of Theorem 3.8 jointly force $\sigma(\Phi_h) = \mathcal{B}$ modulo μ , and the Stone space collapses onto X itself: the abstract completion is the concrete state space. The question left open in [12] — whether prediction, dynamics, and state recovery follow from observation — is answered affirmatively, under faithfulness. What faithfulness means in practice, and whether it can be certified from finite data, remain open.

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